RESULTS AND OUTPUT (SCREENSHOT)

FINAL MODEL RANKING

	Model	Accuracy	Precision (weighted)	Recall (weighted)	F1-score (weighted)	F1-score (macro)	Training time (s)	Testing time (s)
0	Naive Bayes	0.7878	0.7969	0.7878	0.7840	0.7752	0.0818	0.0086
1	SVM	0.7758	0.7752	0.7758	0.7748	0.7671	1.4923	0.0063
2	KNN	0.6866	0.6995	0.6866	0.6884	0.6821	0.0277	2.4150
3	Decision Tree	0.4812	0.5279	0.4812	0.4949	0.4858	26.4963	0.0057

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Best model: Naive Bayes
Weighted F1-score: 0.784
Accuracy: 0.7878
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Naive Bayes
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                          precision    recall  f1-score   support

             alt.atheism       0.73      0.60      0.65       156
           comp.graphics       0.73      0.78      0.76       190
 comp.os.ms-windows.misc       0.74      0.70      0.72       189
comp.sys.ibm.pc.hardware       0.68      0.75      0.71       192
   comp.sys.mac.hardware       0.82      0.78      0.80       185
          comp.windows.x       0.84      0.84      0.84       193
            misc.forsale       0.81      0.74      0.78       189
               rec.autos       0.84      0.84      0.84       187
         rec.motorcycles       0.84      0.78      0.81       194
      rec.sport.baseball       0.93      0.86      0.90       191
        rec.sport.hockey       0.87      0.92      0.90       195
               sci.crypt       0.78      0.82      0.80       193
         sci.electronics       0.82      0.74      0.78       191
                 sci.med       0.90      0.88      0.89       191
               sci.space       0.81      0.84      0.82       190
  soc.religion.christian       0.62      0.93      0.74       195
      talk.politics.guns       0.66      0.87      0.75       177
   talk.politics.mideast       0.89      0.89      0.89       183
      talk.politics.misc       0.78      0.68      0.73       151
      talk.religion.misc       0.84      0.26      0.40       121

                accuracy                           0.79      3653
               macro avg       0.80      0.78      0.78      3653
            weighted avg       0.80      0.79      0.78      3653
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Decision Tree
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                          precision    recall  f1-score   support

            alt.atheism       0.47      0.37      0.42       156
          comp.graphics       0.42      0.47      0.44       190
comp.os.ms-windows.misc       0.51      0.43      0.46       189
comp.sys.ibm.pc.hardware      0.48      0.44      0.46       192
   comp.sys.mac.hardware      0.51      0.44      0.47       185
          comp.windows.x      0.64      0.50      0.56       193
            misc.forsale      0.61      0.49      0.54       189
               rec.autos      0.18      0.61      0.28       187
         rec.motorcycles      0.69      0.53      0.60       194
      rec.sport.baseball      0.63      0.51      0.56       191
        rec.sport.hockey      0.68      0.67      0.67       195
               sci.crypt      0.65      0.58      0.62       193
         sci.electronics      0.38      0.40      0.39       191
                 sci.med      0.60      0.49      0.54       191
               sci.space      0.63      0.56      0.59       190
  soc.religion.christian      0.52      0.45      0.48       195
      talk.politics.guns      0.47      0.46      0.46       177
   talk.politics.mideast      0.78      0.62      0.69       183
      talk.politics.misc      0.36      0.30      0.33       151
      talk.religion.misc      0.19      0.12      0.15       121

                accuracy                           0.48      3653
               macro avg      0.52      0.47      0.49      3653
            weighted avg      0.53      0.48      0.49      3653
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KNN
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                          precision    recall  f1-score   support

            alt.atheism       0.45      0.62      0.52       156
          comp.graphics       0.53      0.73      0.62       190
comp.os.ms-windows.misc       0.56      0.69      0.62       189
comp.sys.ibm.pc.hardware      0.52      0.61      0.56       192
   comp.sys.mac.hardware      0.59      0.64      0.61       185
          comp.windows.x      0.72      0.74      0.73       193
            misc.forsale      0.58      0.49      0.53       189
               rec.autos      0.66      0.73      0.69       187
         rec.motorcycles      0.78      0.69      0.73       194
      rec.sport.baseball      0.85      0.77      0.81       191
        rec.sport.hockey      0.84      0.88      0.86       195
               sci.crypt      0.81      0.78      0.79       193
         sci.electronics      0.73      0.58      0.65       191
                 sci.med      0.84      0.76      0.80       191
               sci.space      0.80      0.68      0.74       190
  soc.religion.christian      0.74      0.74      0.74       195
      talk.politics.guns      0.73      0.67      0.70       177
   talk.politics.mideast      0.87      0.79      0.83       183
      talk.politics.misc      0.70      0.60      0.65       151
      talk.religion.misc      0.62      0.37      0.46       121

                accuracy                           0.69      3653
               macro avg      0.70      0.68      0.68      3653
            weighted avg      0.70      0.69      0.69      3653
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SVM
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	precision	recall	f1-score	support
alt.atheism	0.68	0.63	0.66	156
comp.graphics	0.76	0.75	0.76	190
comp.os.ms-windows.misc	0.74	0.71	0.72	189
comp.sys.ibm.pc.hardware	0.70	0.66	0.68	192
comp.sys.mac.hardware	0.70	0.77	0.74	185
comp.windows.x	0.84	0.80	0.82	193
misc.forsale	0.76	0.75	0.75	189
rec.autos	0.80	0.82	0.81	187
rec.motorcycles	0.79	0.81	0.80	194
rec.sport.baseball	0.89	0.84	0.87	191
rec.sport.hockey	0.90	0.92	0.91	195
sci.crypt	0.88	0.80	0.84	193
sci.electronics	0.72	0.76	0.74	191
sci.med	0.87	0.88	0.88	191
sci.space	0.80	0.83	0.81	190
soc.religion.christian	0.74	0.82	0.78	195
talk.politics.guns	0.70	0.76	0.73	177
talk.politics.mideast	0.88	0.89	0.88	183
talk.politics.misc	0.70	0.71	0.71	151
talk.religion.misc	0.52	0.42	0.47	121
accuracy			0.78	3653
macro avg	0.77	0.77	0.77	3653
weighted avg	0.78	0.78	0.77	3653